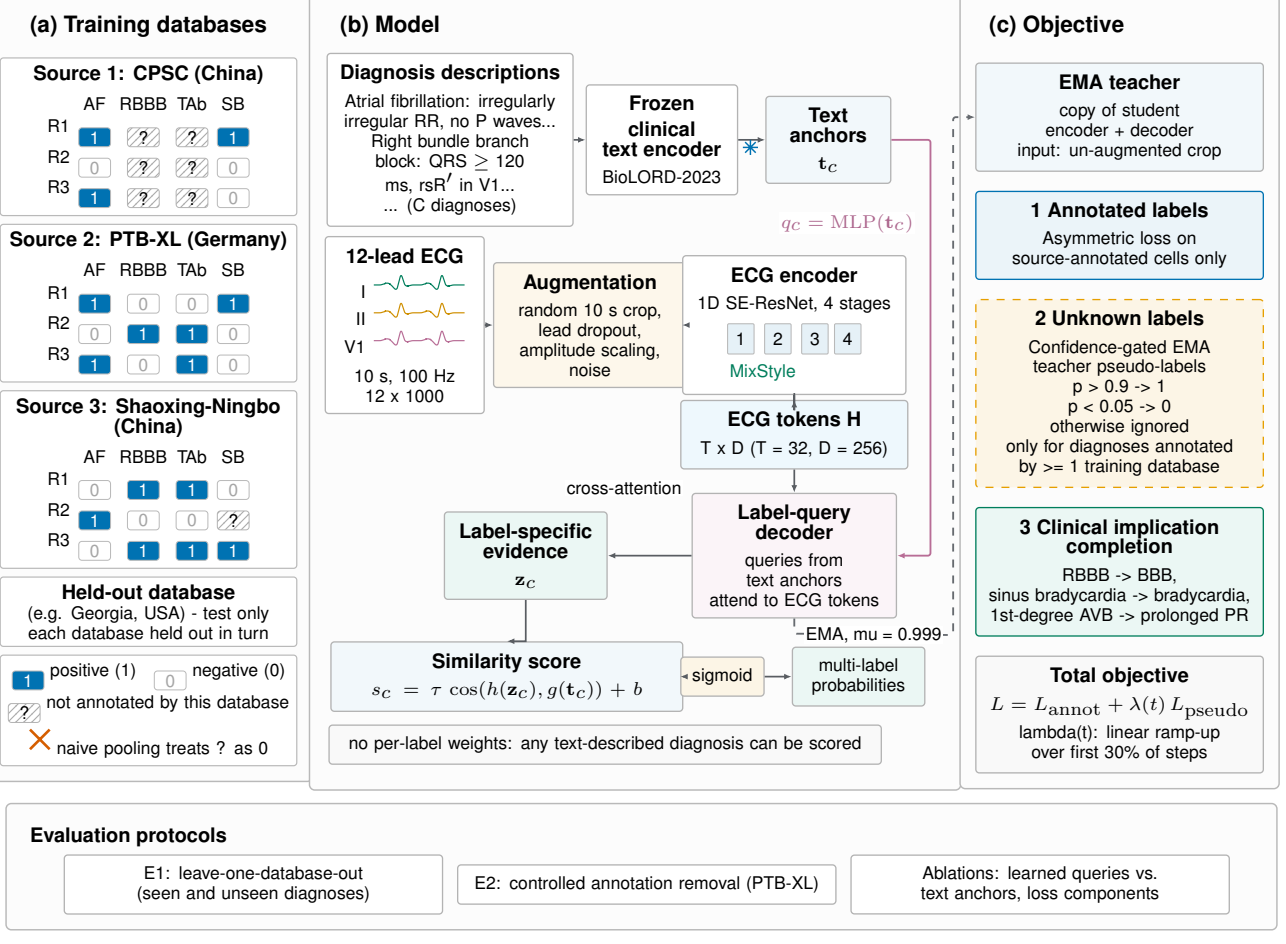

(a) Training databases
Source 1: CPSC (China)
AF RBBB TAb SB
R1 1 ? ? 1
R2 0 ? ? 0
R3 1 ? ? 0
Source 2: PTB-XL (Germany)
AF RBBB TAb SB
R1 1 0 0 1
R2 0 1 1 0
R3 1 0 1 0
Source 3: Shaoxing-Ningbo (China)
AF RBBB TAb SB
R1 0 1 1 0
R2 1 0 0 ?
R3 0 1 1 1
Held-out database
(e.g. Georgia, USA) - test only each database held out in turn
1 positive (1)
0 negative (0)
? not annotated by this database
naive pooling treats ? as 0
(b) Model
Diagnosis descriptions
Atrial fibrillation: irregularly irregular RR, no P waves...
Right bundle branch block: QRS ≥ 120 ms, rsR′ in V1...
... (C diagnoses)
Frozen clinical text encoder
BioLORD-2023
Text anchors
$\mathbf{t}_c$
$q_c = \mathrm{MLP}(\mathbf{t}_c)$
12-lead ECG
I
II
V1
10 s, 100 Hz
12 x 1000
Augmentation
random 10 s crop, lead dropout, amplitude scaling, noise
ECG encoder
1D SE-ResNet, 4 stages
1 2 3 4
MixStyle
ECG tokens H
T x D (T = 32, D = 256)
cross-attention
Label-query decoder
queries from text anchors attend to ECG tokens
Label-specific evidence
$\mathbf{z}_c$
Similarity score
$s_c = \tau \cos(h(\mathbf{z}_c), g(\mathbf{t}_c)) + b$
sigmoid
multi-label probabilities
EMA, mu = 0.999
no per-label weights: any text-described diagnosis can be scored
(c) Objective
EMA teacher
copy of student encoder + decoder
input: un-augmented crop
1 Annotated labels
Asymmetric loss on source-annotated cells only
2 Unknown labels
Confidence-gated EMA teacher pseudo-labels
p > 0.9 -> 1
p < 0.05 -> 0
otherwise ignored
only for diagnoses annotated by >= 1 training database
3 Clinical implication completion
RBBB -> BBB,
sinus bradycardia -> bradycardia,
1st-degree AVB -> prolonged PR
Total objective
$L = L_{\mathrm{annot}} + \lambda(t)\, L_{\mathrm{pseudo}}$
lambda(t): linear ramp-up over first 30% of steps
Evaluation protocols
E1: leave-one-database-out (seen and unseen diagnoses)
E2: controlled annotation removal (PTB-XL)
Ablations: learned queries vs. text anchors, loss components